

6 Chapter 6: Maxwell's Equations for Time-Varying Fields

6.0 Dynamic Fields

$\vec{E}$ and $\vec{D}$ in the static electric model **are not** related to $\vec{H}$ and $\vec{B}$ in the static magnetic model. In addition, static electric and magnetic fields **do not** give rise to waves that propagate and carry energy and information.

In this chapter, we will see that a **changing** magnetic field induces an electric field, and vice versa; that is, they are coupled, and $\vec{E}$ and $\vec{D}$ will be related to $\vec{H}$ and $\vec{B}$, contrary to their lack of interaction in the static model.

6.1 Faraday's Law

To review, in the static case,

$$\nabla \times \vec{E} = 0.$$

Modifying this equation to cover time-varying fields, in a general sense,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}.$$

When $\frac{\partial \vec{B}}{\partial t} = 0$, this relationship simplifies to the static case. Taking the surface integral of both sides of the equation,

$$\int \int_S (\nabla \times \vec{E}) \cdot d\vec{s} = - \int \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}.$$

Using Stoke's theorem to simplify,

$$\oint_C \vec{E} \cdot d\vec{l} = \frac{-d}{dt} \int \int_S \vec{B} \cdot d\vec{s} = \frac{-d}{dt} \Phi.$$

The process described by the above equation is known as **electromagnetic induction**, and the left-hand side can be called the induced voltage, V_{emf} .

Extending to a general case, given N turns in a circuit,

$$V_{\text{emf}} = -N \frac{d\Phi}{dt}.$$

In other words, the voltage induced by a changing magnetic field in a closed circuit is proportional to the rate of change in the magnetic flux and the number of turns N in the circuit. Obviously, the induced voltage then induces a corresponding current.

There are different types of induction. 1) stationary circuit/time-varying $\vec{B} \rightarrow$ transformer induction. This is how transformers operate, by using a changing magnetic field to induce a voltage/current in a stationary circuit. 2) Moving circuit/static $\vec{B} \rightarrow$ motional induction. 3) moving circuit/time-changing $\vec{B} \rightarrow$ general case.

Mathematically,

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tf}} + V_{\text{emf}}^{\text{m}}.$$

6.2 Stationary Loop in a Time-Varying Field

Reviewing each in more detail, starting with transformer induction, recall that in this type of induction, there will be an induced current generated in a closed loop/circuit due to a changing $\vec{B}$ field. Note that the flux produced by the induced current opposes the change in $\vec{B}$ (Lenz's Law).

Imagine a loop of wire (stationary) through which a changing $\vec{B}$ is passing, and it is growing weaker with time. What direction will the induced current flow? Knowing that the $\vec{B}$ field is decreasing, the induced $\vec{B}$

field will *oppose* this change, and increase. Using the right-hand rule, we can find the direction of the induced current. In the opposite case, where $\vec{B}$ increases, the current will flow in the *opposite* direction.

Example time. A circular loop of N turns of conducting wire lies in the x - y plane with its center at the origin of a magnetic field given by $\vec{B} = \hat{\alpha}_z B_0 \cos(\frac{\pi r}{2b}) \sin(\omega t)$, where b is the radius of the loop and ω is the angular frequency. Given this, find the emf induced in the loop.

First, we can identify the type of induction as transformer induction, since the loop is stationary, and $\vec{B}$ is a function of time t . To find emf,

$$V_{\text{emf}} = -N \frac{d\Phi}{dt}.$$

Therefore, we must first find magnetic flux

Φ as a function of our circuit's structure and the magnetic field $\vec{B}$.

Using the formula for magnetic flux,

$$\Phi = \int \int_S \vec{B} \cdot d\vec{s}.$$

In this case, for a single loop, $d\vec{s} = \hat{\alpha}_z r dr d\phi$.

So,

$$\Phi = \int_0^{2\pi} \int_0^b (\hat{\alpha}_z B_0 \cos(\frac{\pi r}{2b}) \sin(\omega t)) \cdot (\hat{\alpha}_z r dr d\phi).$$

Simplifying using integration by parts,

$$\Phi = 2\pi B_0 \sin(\omega t) \int_0^b \cos(\frac{\pi r}{2b}) \cdot r \cdot dr = 2\pi B_0 \sin(\omega t) (\frac{2b^2}{\pi} - \frac{4b^2}{\pi^2}).$$

Taking its time derivative, only the sin function matters, so

$$\frac{d\Phi}{dt} = 2\pi B_0 (\frac{2b^2}{\pi} - \frac{4b^2}{\pi^2}) \omega \cos(\omega t)$$

and so

$$V_{\text{emf}} = -2N\pi B_0 (\frac{2b^2}{\pi} - \frac{4b^2}{\pi^2}) \omega \cos(\omega t) \text{ V}.$$

Another example problem: an inductor is formed by winding N turns of a thin conducting wire into a circular loop of radius a . The inductor loop is in the x - y plane with its center at the origin, and connected to a resistor R . In the presence of a magnetic field $\vec{B} = B_0(2\hat{\alpha}_y + 3\hat{\alpha}_z) \sin(\omega t)$. Find

- a) the magnetic flux linking a single turn of the inductor. In this case, the changing magnetic field has both y and z components, and not all of it will end up as magnetic flux Φ as a result. $d\vec{s}$ is the same as in the prior problem, $d\vec{s} = \hat{\alpha}_z (r dr d\phi)$. Finding magnetic flux using the same integral setup as before,

$$\Phi = \int_0^{2\pi} \int_0^a B_0(2\hat{\alpha}_y + 3\hat{\alpha}_z) \sin(\omega t) \cdot (\hat{\alpha}_z (r dr d\phi)).$$

Simplifying,

$$\Phi = 3B_0\pi a^2 \sin(\omega t).$$

- b) the emf given that $N = 10$, $B_0 = a2T$, $a = 10 \text{ cm}$, and $\omega = 10^3 \frac{\text{rad}}{\text{s}}$.

Knowing that

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = -N \cdot 3\pi a^2 B_0 \omega \cos(\omega t),$$

substituting values,

$$V_{\text{emf}} = -188.5 \cos(10^3 t) \text{ V}.$$

- c) the direction of the induced current at $t = 0$. Recall that induced current always opposes the direction of change in the magnetic flux; therefore, the question is: is magnetic flux Φ increasing or decreasing at $t = 0$?

We may evaluate the derivative $\frac{d\Phi}{dt}$ at $t = 0$ to determine whether Φ is increasing or decreasing. At $t = 0$, the derivative is positive (greater than zero), and so we may assume that Φ is increasing at $t = 0$. Therefore, flux is increasing in the z direction at $t = 0$, and so the current will flow in the clockwise direction (right-hand rule for the negative z direction).

6.3 The Ideal Transformer

An ideal transformer has two sides, with loops of wire (N turns) wrapped around each side; generally, one side is connected to an AC source, and the other to a load. Using Faraday's Law,

$$V = -N \frac{d\Phi}{dt},$$

we can generalize to the case of an ideal transformer, where

$$\frac{V_1}{V_2} = \frac{N_1}{N_2},$$

while

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}.$$

In addition, $R_{\text{in}} = R_L \frac{N_1^2}{N_2^2}$ and $Z_{\text{in}} = Z_L \frac{N_1^2}{N_2^2}$.

6.4 Moving Conductor in a Static Magnetic Field

As an example, a conducting wire of length l moves across a static magnetic field $\vec{B} = \hat{\alpha}_z B_0$ with velocity $\vec{u} = \hat{\alpha}_x u$.

Recall that

$$\vec{F}_m = q(\vec{u} \times \vec{B}).$$

Therefore,

$$\vec{E}_m = \frac{\vec{F}_m}{q} = \vec{u} \times \vec{B}.$$

This will induce a potential difference across the two ends of the wire, and

$$V_{\text{emf}} = \int_2^1 \vec{E}_m \cdot d\vec{l} = \int_2^1 (\vec{u} \times \vec{B}) \cdot d\vec{l} = \int (\hat{\alpha}_x u x \hat{\alpha}_z B_0) \cdot \hat{\alpha}_y dy = -u B_0 l.$$

In general, if any segment of a closed circuit moves with a velocity $\vec{u}$ across a static magnetic field $\vec{B}$, then the induced motional emf is

$$V_{\text{emf}}^m = \oint (\vec{u} \times \vec{B}) \cdot d\vec{l}.$$

As an example, a metal bar slides over a pair of conducting rails in a uniform magnetic field $\vec{B} = \hat{\alpha}_z B_0$ with a velocity $\vec{u} = \hat{\alpha}_x u$.

- a) Determine the open circuit voltage V_0 that appears across terminals 1 and 2.

First, determine the direction of the current; current will flow from protons to electrons, so clockwise. Then, find the voltage,

$$\begin{aligned} V_0 &= \int (\vec{u} \times \vec{B}) \cdot d\vec{l} \\ &= -u B_0 h \text{ V} \end{aligned}$$

- b) Assuming that a resistance R is connected between the terminals, find the electric power dissipated in the resistor.

First,

$$I = \frac{uB_0h}{R},$$

and so

$$P = I^2R.$$

As another example, the rectangular loop shown below is located in the x - y plane and moves away from the origin with velocity $\vec{u} = 5\hat{\alpha}_y \frac{\text{m}}{\text{s}}$ in a magnetic field given by $\vec{B}(y) = \hat{\alpha}_z 0.2e^{0.1y} \text{ T}$. If $R = 5\Omega$, find the current I at the instant that the loop sides are at $y_1 = 2$ and $y_2 = 2.5$ m.

Recall the formula for motional emf,

$$V_{\text{emf}}^m = \oint_C (\vec{u} \times \vec{B}) \cdot d\vec{l}.$$

Since $\vec{u} \times \vec{B}$ is along $\hat{\alpha}_x$, voltages are induced across only the sides oriented along $\hat{\alpha}_x$ (1 and 2, and 3 and 4).